PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 3 · Version A · Solutions

20 questions · 90 minutes · original practice exam, not an official exam

ANSWER KEY

1	**a**	2	**a**	3	**c**	4	**a**	5	**b**	6	**c**	7	**a**
8	**a**	9	**c**	10	**b**	11	**e**	12	**b**	13	**c**	14	**b**
15	**e**	16	**d**	17	**d**	18	**c**	19	**c**	20	**a**		

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A mechanic pulls on the end of a 0.25 m wrench with an 80 N force directed at 60° to the wrench handle.

1) What is the magnitude of the torque on the bolt?

Answer: (a) 17.3 N·m

$\tau = rF\sin\varphi = (0.25)(80)(\sin 60°) = 17.3$ N·m.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1.0 kg block hangs from a light rope wrapped around a pulley that is a uniform solid disk (M = 2.0 kg, R = 0.10 m). The pulley turns without friction, and the block is released from rest.

2) What is the acceleration of the block?

Answer: (a) 4.91 m/s²

$mg - T = ma$ and $TR = \frac{1}{2}MR^2(a/R) \rightarrow a = mg/(m + M/2) = 4.91$ m/s².

3) What is the tension in the rope?

Answer: (c) 4.91 N

$T = m(g - a) = 4.91$ N.

4) If the pulley were a thin ring with the same mass and radius, the block's acceleration would be

Answer: (a) smaller.

A ring has a larger moment of inertia, so it resists turning more.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A solid sphere (m = 2.0 kg, R = 0.10 m) rolls without slipping from rest down an incline. It descends a vertical height of 1.5 m.

5) What is the speed of the sphere at the bottom?

Answer: (b) 4.58 m/s

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}(\frac{2}{5}mR^2)(v/R)^2 = (7/10)mv^2 \rightarrow v = \sqrt{(10gh/7)} = 4.58$ m/s.

6) What fraction of the sphere's kinetic energy at the bottom is rotational?

Answer: (c) 28.6%

$K_{rot}/K = (\frac{2}{5})/(1 + \frac{2}{5}) = 2/7 = 28.6\%$.

7) How much work does static friction do on the sphere while it rolls?

Answer: (a) Zero

The contact point is momentarily at rest, so static friction does no work.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A uniform 6.0 m plank of mass 30 kg rests on supports at its two ends. A 70 kg person stands 1.5 m from the left end.

8) What upward force does the right support exert?

Answer: (a) 319 N

Torques about the left end: $R(6) = (30g)(3) + (70g)(1.5)$, R = 319 N.

9) What upward force does the left support exert?

Answer: (c) 662 N

$\Sigma F = 0$: $L = 100g - R = 662$ N.

10) As the person walks toward the right end, the force from the left support

Answer: (b) decreases.

More of the weight shifts onto the right support.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A figure skater spins at 2.0 rev/s with her arms out ($I = 4.0$ kg·m²). She pulls her arms in, reducing her moment of inertia to 1.6 kg·m².

11) What is her new rate of spin?

Answer: (e) 5 rev/s

$I_1\omega_1 = I_2\omega_2$: $\omega_2 = (4.0)(2.0)/1.6 = 5$ rev/s.

12) Her rotational kinetic energy

Answer: (b) increases.

$K = L^2/2I$. L is fixed and I drops, so K rises. Her muscles do the work.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 30 kg child runs at 4.0 m/s along a line tangent to the edge of a merry-go-round and jumps on at its edge. The merry-go-round is a uniform disk of mass 120 kg and radius 2.0 m, initially at rest.

13) What is the angular speed of the merry-go-round just after the child lands?

Answer: (c) 0.667 rad/s

$L = mvR = 240\ kg{\cdot}m^2/s$; $I = \frac{1}{2}(120)(2^2) + 30(2^2) = 360\ kg{\cdot}m^2$; $\omega = 0.667$ rad/s.

14) Compared with before the jump, the total kinetic energy afterward is

Answer: (b) less.

This is an inelastic "collision": L is conserved but kinetic energy is not.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A constant torque of 6.0 N·m acts on a wheel ($I = 1.5\ kg{\cdot}m^2$) that starts at rest.

15) What is the angular momentum of the wheel after 4.0 s?

Answer: (e) 24 kg·m²/s

$\Delta L = \tau\Delta t = (6.0)(4.0) = 24\ kg{\cdot}m^2/s$.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.80 kg block on a horizontal frictionless surface is attached to a spring with k = 200 N/m. It oscillates with an amplitude of 0.10 m.

16) What is the period of the oscillation?

Answer: (d) 0.397 s

$\omega = \sqrt{(k/m)} = 15.8$ rad/s; $T = 2\pi/\omega = 0.397$ s.

17) What is the maximum speed of the block?

Answer: (d) 1.58 m/s

$v_{max} = A\omega = (0.10)(15.8) = 1.58$ m/s.

18) When the block is at x = A/2, its speed is what fraction of v_{max}?

Answer: (c) 0.87

$\frac{1}{2}kA^2 = \frac{1}{2}k(A/2)^2 + \frac{1}{2}mv^2 \rightarrow v = (\sqrt{3}/2)v_{max} \approx 0.87\ v_{max}$.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A simple pendulum has a period of 2.0 s on Earth. It is taken to a planet where g is one quarter of its value on Earth.

19) What is its period on the planet?

Answer: (c) 4 s

$T \propto 1/\sqrt{g}$; g/4 doubles the period to 4.0 s.

20) If the amplitude of the swing were doubled (still small), the period would

Answer: (a) stay the same.

For small oscillations the period does not depend on amplitude.
